1 Spherical harmonics

$$\frac{1}{l}=\begin{cases}\frac{1}{r}\left[1+\left(\frac{r'}{r}\right)^2-2\frac{r'}{r}\cos\psi\right]^{-1/2}=\sum_{n=0}^{\infty}\frac{r'^n}{r^{n+1}}P_n(\cos\psi) & r>r'\\ \frac{1}{r'}\left[1+\left(\frac{r}{r'}\right)^2-2\frac{r}{r'}\cos\psi\right]^{-1/2}=\sum_{n=0}^{\infty}\frac{r^n}{r'^{n+1}}P_n(\cos\psi) & r<r'\end{cases}. \tag{1}$$

$$\mathcal{H}_n^k(\theta,\lambda)=\begin{cases}P_n^{|k|}(\cos\theta)\cos|k|\lambda, & k\geq 0;\\ P_n^{|k|}(\cos\theta)\sin|k|\lambda, & k<0.\end{cases} \tag{2}$$

$$P_n(\cos\psi)=\sum_{k=-n}^{n}\frac{2}{1+\delta_k}\frac{(n-|k|)!}{(n+|k|)!}\mathcal{H}_n^k(\theta,\lambda)\mathcal{H}_n^k(\theta',\lambda'). \tag{3}$$

$$\int_\omega \mathcal{H}_n^k(\theta,\lambda)\mathcal{H}_m^l(\theta,\lambda)dw=\delta_{mn}\delta_{kl}\frac{2\pi(1+\delta_k)}{2n+1}\frac{(n+|k|)!}{(n-|k|)!}. \tag{4}$$

2 Spherical Earth

$$\mu(\theta,\lambda,t)=\sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\mu_n^k(t)\mathcal{H}_n^k(\theta,\lambda). \tag{5}$$

$$\phi(r,\theta,\lambda,t)=\begin{cases}\sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\frac{4\pi GR^2}{2n+1}\mu_n^k(t)\frac{R^n}{r^{n+1}}\mathcal{H}_n^k(\theta,\lambda), & r\geq R;\\ \sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\frac{4\pi GR^2}{2n+1}\mu_n^k(t)\frac{r^n}{R^{n+1}}\mathcal{H}_n^k(\theta,\lambda), & r\leq R.\end{cases} \tag{6}$$

3 Aspherical Earth

In this case, the radial distance of a point on the Earth's surface is position dependent. We express it as

$$r_S(\theta,\lambda)=R+h(\theta,\lambda)=R[1+\epsilon(\theta,\lambda)]. \tag{7}$$

This implies that the actual surface of the Earth is h above the sphere of radius R, with $\epsilon=h/R$ as a relative measure of the height. We define the surface density at a point as the mass per unit area on the equipotential surface of gravity through the point, which conforms with the mass of a vertical water column of unit bottom area. We still express the time-variable surface mass density as a spherical harmonic series,

$$\mu(\theta,\lambda,t)=\sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\mu_n^k(t)\mathcal{H}_n^k(\theta,\lambda). \tag{8}$$

Based on the definition of $\mu(\theta,\lambda,t)$, the mass of an infinitesimal area at (θ',λ') is

$$dm(\theta',\lambda',t)=\frac{r_S^2(\theta',\lambda')\mu(\theta',\lambda',t)\sin\theta' d\theta' d\lambda'}{\cos(-\hat{e}_r,\hat{e}_g)}, \tag{9}$$

where $(-\hat{e}_r,\hat{e}_g)$ is the angle between the directions of the geocenter and the plumb line. The difference between $\cos(-\hat{e}_r,\hat{e}_g)$ and 1 is of second order of magnitude of the angle. Even the effect of the Earth's flattening is on the order of 6×10^{-6}. Hence, accurate to the first order of magnitude of the asphericity, we can make the approximation $\cos(-\hat{e}_r,\hat{e}_g)=1$, which leads to the approximation

$$dm(\theta',\lambda',t)=r_S^2(\theta',\lambda')\mu(\theta',\lambda',t)\sin\theta' d\theta' d\lambda'. \tag{10}$$

The gravitational potential of this infinitesimal mass at (r, θ, λ) is

$$d\phi(r,\theta,\lambda,t) = G\frac{r_S^2(\theta',\lambda')}{l[(r,\theta,\lambda),(r_S,\theta',\lambda')]}\mu(\theta',\lambda',t)\sin\theta' d\theta' d\lambda' \,, \tag{11}$$

where l is the distance between the two points (r, θ, λ) and (r_S, θ', λ'). The total gravitational potential is thus

$$\phi(r,\theta,\lambda,t) = G\int_\omega \frac{r_S^2(\theta',\lambda')}{l[(r,\theta,\lambda),(r_S,\theta',\lambda')]}\mu(\theta',\lambda',t) d\omega \,, \tag{12}$$

where ω is the unit sphere with $d\omega = \sin\theta' d\theta' d\lambda'$.

From now on, we keep implicit the variables that the functions depend on. Denote the geocentric angle between the two points (r, θ, λ) and (r_S, θ', λ') as ψ. We have

$$l = \left(r^2 + r_S^2 - 2 r r_S \cos\psi\right)^{1/2} \,, \tag{13}$$

$$\cos\psi = \cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\lambda - \lambda') \,. \tag{14}$$

According to (1), we express $1/l$ as

$$\frac{1}{l} = \begin{cases} \displaystyle\sum_{p=0}^{\infty} \frac{r_S^p}{r^{p+1}} P_p(\cos\psi) \,, & r > r_S ; \\ \displaystyle\sum_{p=0}^{\infty} \frac{r^p}{r_S^{p+1}} P_p(\cos\psi) \,, & r < r_S \,. \end{cases} \tag{15}$$

Replace θ and λ by θ' and λ', respectively, in (8), and then substitute it into (12). By further using (15), we obtain

$$\phi = \begin{cases} \displaystyle G\sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\sum_{p=0}^{\infty}\mu_n^k \int_\omega \frac{r_S^{p+2}}{r^{p+1}} P_p(\cos\psi)\mathcal{H}_n^k(\theta',\lambda') d\omega \,, & r > \max(r_S) ; \\ \displaystyle G\sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\sum_{p=0}^{\infty}\mu_n^k \int_\omega \frac{r^p}{r_S^{p-1}} P_p(\cos\psi)\mathcal{H}_n^k(\theta',\lambda') d\omega \,, & r < \min(r_S) ; \end{cases} \tag{16}$$

Note the difference in the domain of convergence between (15) and (16), the former being for a single point on the Earth's surface, while the latter for any point. We write the addition theorem (3) as

$$P_p(\cos\psi) = \sum_{q=-p}^{p} \frac{2}{1+\delta_q} \frac{(p-|q|)!}{(p+|q|)!} \mathcal{H}_p^q(\theta,\lambda)\mathcal{H}_p^q(\theta',\lambda') \,. \tag{17}$$

It can be used to write (16) as

$$\phi = \begin{cases} \displaystyle G\sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\mu_n^k \int_\omega \frac{r_S^{p+2}}{r^{p+1}} \frac{2}{1+\delta_q} \frac{(p-|q|)!}{(p+|q|)!} \mathcal{H}_p^q(\theta,\lambda)\mathcal{H}_p^q(\theta',\lambda')\mathcal{H}_n^k(\theta',\lambda') d\omega \,, & r > \max(r_S) ; \\ \displaystyle G\sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\mu_n^k \int_\omega \frac{r^p}{r_S^{p-1}} \frac{2}{1+\delta_q} \frac{(p-|q|)!}{(p+|q|)!} \mathcal{H}_p^q(\theta,\lambda)\mathcal{H}_p^q(\theta',\lambda')\mathcal{H}_n^k(\theta',\lambda') d\omega \,, & r < \min(r_S) ; \end{cases} \tag{18}$$

$$\phi = \begin{cases} \displaystyle GR^2\sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\mu_n^k \int_\omega \frac{r_S^{p+2}}{R^{p+2}} \frac{R^p}{r^{p+1}} \frac{2}{1+\delta_q} \frac{(p-|q|)!}{(p+|q|)!} \mathcal{H}_p^q(\theta,\lambda)\mathcal{H}_p^q(\theta',\lambda')\mathcal{H}_n^k(\theta',\lambda') d\omega \,, & r > \max(r_S) ; \\ \displaystyle GR^2\sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\mu_n^k \int_\omega \frac{R^{p-1}}{r_S^{p-1}} \frac{r^p}{R^{p+1}} \frac{2}{1+\delta_q} \frac{(p-|q|)!}{(p+|q|)!} \mathcal{H}_p^q(\theta,\lambda)\mathcal{H}_p^q(\theta',\lambda')\mathcal{H}_n^k(\theta',\lambda') d\omega \,, & r < \min(r_S) ; \end{cases} \tag{19}$$

$$\phi = \begin{cases} \displaystyle GR^2\sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\mu_n^k I_{nkpq}^{(e)} \frac{R^p}{r^{p+1}} \mathcal{H}_p^q(\theta,\lambda) \,, & r > \max(r_S) ; \\ \displaystyle GR^2\sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\mu_n^k I_{nkpq}^{(i)} \frac{r^p}{R^{p+1}} \mathcal{H}_p^q(\theta,\lambda) \,, & r < \min(r_S) ; \end{cases} \tag{20}$$

where $I^{(e)}_{nkpq}$ and $I^{(i)}_{nkpq}$ are defined as

$$\left\{\begin{aligned} I^{(e)}_{nkpq} &= \int_\omega \left(\frac{r_S}{R}\right)^{p+2} \frac{2}{1+\delta_q}\frac{(p-|q|)!}{(p+|q|)!}\mathcal{H}^q_p(\theta',\lambda')\mathcal{H}^k_n(\theta',\lambda')d\omega\,,\\ I^{(i)}_{nkpq} &= \int_\omega \left(\frac{R}{r_S}\right)^{p-1} \frac{2}{1+\delta_q}\frac{(p-|q|)!}{(p+|q|)!}\mathcal{H}^q_p(\theta',\lambda')\mathcal{H}^k_n(\theta',\lambda')d\omega\,. \end{aligned}\right. \tag{21}$$

According to (7), we make the arrangements of

$$\left(\frac{r_S}{R}\right)^{p+2} = 1+\left[(1+\epsilon)^{p+2}-1\right]\,, \qquad \left(\frac{R}{r_S}\right)^{p-1} = 1+\left[\frac{1}{(1+\epsilon)^{p-1}}-1\right]. \tag{22}$$

We further define $\delta k^{h_e}_{mlpq}$ and $\delta k^{h_i}_{mlpq}$ through

$$\left\{\begin{aligned} \delta k^{h_e}_{pq} &= \left[(1+\epsilon)^{p+2}-1\right]\frac{2}{1+\delta_q}\frac{(p-|q|)!}{(p+|q|)!}\mathcal{H}^q_p(\theta',\lambda') = \sum_{m=0}^{\infty}\sum_{l=-m}^{m}\delta k^{h_e}_{mlpq}\frac{2}{1+\delta_l}\frac{(m-|l|)!}{(m+|l|)!}\mathcal{H}^l_m(\theta',\lambda')\,,\\ \delta k^{h_i}_{pq} &= \left[\frac{1}{(1+\epsilon)^{p-1}}-1\right]\frac{2}{1+\delta_q}\frac{(p-|q|)!}{(p+|q|)!}\mathcal{H}^q_p(\theta',\lambda') = \sum_{m=0}^{\infty}\sum_{l=-m}^{m}\delta k^{h_i}_{mlpq}\frac{2}{1+\delta_l}\frac{(m-|l|)!}{(m+|l|)!}\mathcal{H}^l_m(\theta',\lambda')\,. \end{aligned}\right. \tag{23}$$

As $\delta k^{h_e}_{pq}$ and $\delta k^{h_i}_{pq}$ are known, $\delta k^{h_e}_{mlpq}$ and $\delta k^{h_i}_{mlpq}$ can be computed by expressing $\delta k^{h_e}_{pq}$ and $\delta k^{h_i}_{pq}$ in spherical harmonic series. The substitution of (22) and (23) into (21) leads to

$$\begin{aligned} I^{(e)}_{nkpq} =& \frac{2}{1+\delta_q}\frac{(p-|q|)!}{(p+|q|)!}\int_\omega \mathcal{H}^q_p(\theta',\lambda')\mathcal{H}^k_n(\theta',\lambda')d\omega\\ &+\sum_{m=0}^{\infty}\sum_{l=-m}^{m}\delta k^{h_e}_{mlpq}\frac{2}{1+\delta_l}\frac{(m-|l|)!}{(m+|l|)!}\int_\omega \mathcal{H}^l_m(\theta',\lambda')\mathcal{H}^k_n(\theta',\lambda')d\omega\,, \end{aligned} \tag{24a}$$

$$\begin{aligned} I^{(i)}_{nkpq} =& \frac{2}{1+\delta_q}\frac{(p-|q|)!}{(p+|q|)!}\int_\omega \mathcal{H}^q_p(\theta',\lambda')\mathcal{H}^k_n(\theta',\lambda')d\omega\\ &+\sum_{m=0}^{\infty}\sum_{l=-m}^{m}\delta k^{h_i}_{mlpq}\frac{2}{1+\delta_l}\frac{(m-|l|)!}{(m+|l|)!}\int_\omega \mathcal{H}^l_m(\theta',\lambda')\mathcal{H}^k_n(\theta',\lambda')d\omega\,. \end{aligned} \tag{24b}$$

These can be simplified to, according to the orthogonality (4),

$$\left\{\begin{aligned} I^{(e)}_{nkpq} &= \frac{4\pi}{2n+1}\delta_{pn}\delta_{qk} + \frac{4\pi}{2n+1}\delta k^{h_e}_{nkpq}\,,\\ I^{(i)}_{nkpq} &= \frac{4\pi}{2n+1}\delta_{pn}\delta_{qk} + \frac{4\pi}{2n+1}\delta k^{h_i}_{nkpq}\,. \end{aligned}\right. \tag{25}$$

Finally the result of ϕ is obtained by substituting (25) into (20). We express it as

$$\phi = \phi_0 + \delta\phi = \left\{\begin{aligned} &\phi^{(e)} = \phi^{(e)}_0 + \delta\phi^{(e)}\,, \quad r \geq R;\\ &\phi^{(i)} = \phi^{(i)}_0 + \delta\phi^{(i)}\,, \quad r \leq R; \end{aligned}\right. \tag{26}$$

where

$$\left\{\begin{aligned} \phi^{(e)}_0 &= \sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\frac{4\pi GR^2}{2n+1}\mu^k_n\frac{R^n}{r^{n+1}}\mathcal{H}^k_n(\theta,\lambda)\,,\\ \phi^{(i)}_0 &= \sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\frac{4\pi GR^2}{2n+1}\mu^k_n\frac{r^n}{R^{n+1}}\mathcal{H}^k_n(\theta,\lambda)\,, \end{aligned}\right. \tag{27}$$

$$\left\{\begin{aligned} \delta\phi^{(e)} &= \sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\left[\frac{4\pi GR^2}{2n+1}\mu^k_n\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\delta k^{h_e}_{nkpq}\frac{R^p}{r^{p+1}}\mathcal{H}^q_p(\theta,\lambda)\right],\\ \delta\phi^{(i)} &= \sum_{n=0}^{\infty}\sum_{k=-n}^{k=n}\left[\frac{4\pi GR^2}{2n+1}\mu^k_n\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\delta k^{h_i}_{nkpq}\frac{r^p}{R^{p+1}}\mathcal{H}^q_p(\theta,\lambda)\right]. \end{aligned}\right. \tag{28}$$

We see that $\phi_0^{(e)}$ and $\phi_0^{(i)}$ are already given in (6) for the spherical Earth. In this regards, $\delta\phi^{(e)}$ and $\delta\phi^{(i)}$ are corrections due to the asphericity, i.e., the deviation from sphericity.

4 Practical Formulae

In view of practical applications, we discuss only the external case in terms of real spherical harmonics. The GRACE data are provided in terms of the fully normalized Legendre function and spherical harmonics.

$$s = \sum_{n=0}^{\infty}\sum_{k=-n}^{n} s_n^k \mathcal{H}_n^k(\theta,\lambda)\,; \tag{29}$$

$$\delta k_{pq}^{h_e} = \left[(1+\epsilon)^{p+2}-1\right]\frac{2}{1+\delta_q}\frac{(p-|q|)!}{(p+|q|)!}\mathcal{H}_p^q(\theta',\lambda') \tag{30}$$

$$\delta k_{pq}^{h_e} = \sum_{m=0}^{\infty}\sum_{l=-m}^{m}\delta k_{mlpq}^{h_e}\frac{2}{1+\delta_l}\frac{(m-|l|)!}{(m+|l|)!}\mathcal{H}_m^l(\theta',\lambda') \tag{31}$$

$$\phi_0^{(e)} = Rg_R\sum_{n=0}^{\infty}\sum_{k=-n}^{n}\frac{4\pi G}{(2n+1)g_R}s_n^k\left(\frac{R}{r}\right)^{n+1}\mathcal{H}_n^k(\theta,\lambda) \tag{32}$$

$$\delta\phi^{(e)} = Rg_R\sum_{n=0}^{\infty}\sum_{k=-n}^{n}\left[\frac{4\pi G}{(2n+1)g_R}s_n^k\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\delta k_{nkpq}^{h_e}\left(\frac{R}{r}\right)^{p+1}\mathcal{H}_p^q(\theta,\lambda)\right], \tag{33}$$

$$\phi = Rg_R\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\phi_p^q\left(\frac{R}{r}\right)^{p+1}\mathcal{H}_p^q(\theta,\lambda)\,. \tag{34}$$

$$\phi_p^q = \frac{4\pi G}{g_R}\left[\frac{1}{2p+1}s_p^q + \sum_{n=0}^{\infty}\sum_{k=-n}^{n}\frac{1}{2n+1}s_n^k\delta k_{nkpq}^{h_e}\right]. \tag{35}$$

$$s_p^q = \frac{(2p+1)g_R}{4\pi G}\left[\phi_p^q - \sum_{n=0}^{\infty}\sum_{k=-n}^{n}\phi_n^k\delta k_{nkpq}^{h_e}\right]. \tag{36}$$

$$\bar{P}_n^k(x) = \left[\frac{2(2n+1)}{1+\delta_k}\frac{(n-k)!}{(n+k)!}\right]^{1/2}P_n^k(x) \tag{37}$$

$$\bar{\mathcal{H}}_n^k(\theta,\lambda) = \begin{cases}\bar{P}_n^k(\cos\theta)\cos k\lambda\,, & k\ge 0\,;\\ \bar{P}_n^{|k|}(\cos\theta)\sin|k|\lambda\,, & k<0\,;\end{cases} \tag{38}$$

$$s = \sum_{n=0}^{\infty}\sum_{k=-n}^{n}\left[\frac{1+\delta_k}{2(2n+1)}\frac{(n+|k|)!}{(n-|k|)!}\right]^{1/2}s_n^k\bar{\mathcal{H}}_n^k(\theta,\lambda)\,; \tag{39}$$

$$\delta k_{pq}^{h_e} = \left[(1+\epsilon)^{p+2}-1\right]\frac{2}{1+\delta_q}\frac{(p-|q|)!}{(p+|q|)!}\left[\frac{1+\delta_q}{2(2p+1)}\frac{(p+|q|)!}{(p-|q|)!}\right]^{1/2}\bar{\mathcal{H}}_p^q(\theta',\lambda') \tag{40}$$

$$\delta k_{pq}^{h_e} = \sum_{m=0}^{\infty}\sum_{l=-m}^{m}\delta k_{mlpq}^{h_e}\frac{2}{1+\delta_l}\frac{(m-|l|)!}{(m+|l|)!}\left[\frac{1+\delta_l}{2(2m+1)}\frac{(m+|l|)!}{(m-|l|)!}\right]^{1/2}\bar{\mathcal{H}}_m^l(\theta',\lambda') \tag{41}$$

$$\phi_0^{(e)} = Rg_R\sum_{n=0}^{\infty}\sum_{k=-n}^{n}\frac{4\pi G}{(2n+1)g_R}\left[\frac{1+\delta_k}{2(2n+1)}\frac{(n+|k|)!}{(n-|k|)!}\right]^{1/2}s_n^k\left(\frac{R}{r}\right)^{n+1}\bar{\mathcal{H}}_n^k(\theta,\lambda) \tag{42}$$

$$\delta\phi^{(e)} = Rg_R\sum_{n=0}^{\infty}\sum_{k=-n}^{n}\left\{\frac{4\pi G}{(2n+1)g_R}s_n^k\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\left[\frac{1+\delta_q}{2(2p+1)}\frac{(p+|q|)!}{(p-|q|)!}\right]^{1/2}\delta k_{nkpq}^{h_e}\left(\frac{R}{r}\right)^{p+1}\bar{\mathcal{H}}_p^q(\theta,\lambda)\right\}, \tag{43}$$

$$\phi = Rg_R \sum_{p=0}^{\infty} \sum_{q=-p}^{p} \left[\frac{1+\delta_q}{2(2p+1)} \frac{(p+|q|)!}{(p-|q|)!} \right]^{1/2} \phi_p^q \left(\frac{R}{r} \right)^{p+1} \bar{\mathcal{H}}_p^q(\theta, \lambda) . \tag{44}$$

$$\phi_p^q = \frac{4\pi G}{g_R} \left[\frac{1}{2p+1} s_p^q + \sum_{n=0}^{\infty} \sum_{k=-n}^{n} \frac{1}{2n+1} s_n^k \delta k_{nkpq}^{h_e} \right] . \tag{45}$$

$$s_p^q = \frac{(2p+1) g_R}{4\pi G} \left[\phi_p^q - \sum_{n=0}^{\infty} \sum_{k=-n}^{n} \phi_n^k \delta k_{nkpq}^{h_e} \right] . \tag{46}$$

$$s_n^k = \left[\frac{2(2n+1)}{1+\delta_k} \frac{(n-|k|)!}{(n+|k|)!} \right]^{1/2} \bar{s}_n^k \tag{47}$$

$$s = \sum_{n=0}^{\infty} \sum_{k=-n}^{n} \bar{s}_n^k \bar{\mathcal{H}}_n^k(\theta, \lambda) ; \tag{48}$$

$$\delta k_{pq}^{h_e} = \left[(1+\epsilon)^{p+2} - 1 \right] \frac{2}{1+\delta_q} \frac{(p-|q|)!}{(p+|q|)!} \left[\frac{1+\delta_q}{2(2p+1)} \frac{(p+|q|)!}{(p-|q|)!} \right]^{1/2} \bar{\mathcal{H}}_p^q(\theta', \lambda') \tag{49}$$

$$\delta k_{pq}^{h_e} = \sum_{m=0}^{\infty} \sum_{l=-m}^{m} \delta k_{mlpq}^{h_e} \frac{2}{1+\delta_l} \frac{(m-|l|)!}{(m+|l|)!} \left[\frac{1+\delta_l}{2(2m+1)} \frac{(m+|l|)!}{(m-|l|)!} \right]^{1/2} \bar{\mathcal{H}}_m^l(\theta', \lambda') \tag{50}$$

$$\phi_0^{(e)} = Rg_R \sum_{n=0}^{\infty} \sum_{k=-n}^{n} \frac{4\pi G}{(2n+1) g_R} \bar{s}_n^k \left(\frac{R}{r} \right)^{n+1} \bar{\mathcal{H}}_n^k(\theta, \lambda) \tag{51}$$

$$\begin{aligned} \delta\phi^{(e)} = Rg_R \sum_{n=0}^{\infty} \sum_{k=-n}^{n} \Bigg\{ & \frac{4\pi G}{(2n+1) g_R} \left[\frac{2(2n+1)}{1+\delta_k} \frac{(n-|k|)!}{(n+|k|)!} \right]^{1/2} \bar{s}_n^k \\ & \times \sum_{p=0}^{\infty} \sum_{q=-p}^{p} \left[\frac{1+\delta_q}{2(2p+1)} \frac{(p+|q|)!}{(p-|q|)!} \right]^{1/2} \delta k_{nkpq}^{h_e} \left(\frac{R}{r} \right)^{p+1} \bar{\mathcal{H}}_p^q(\theta, \lambda) \Bigg\} , \end{aligned} \tag{52}$$

$$\phi = Rg_R \sum_{p=0}^{\infty} \sum_{q=-p}^{p} \bar{\phi}_p^q \left(\frac{R}{r} \right)^{p+1} \bar{\mathcal{H}}_p^q(\theta, \lambda) . \tag{53}$$

$$\delta k_{nkpq}^{h_e} = \left[\frac{1+\delta_k}{2(2n+1)} \frac{(n+|k|)!}{(n-|k|)!} \right]^{1/2} \left[\frac{2(2p+1)}{1+\delta_k} \frac{(p-|q|)!}{(p+|q|)!} \right]^{1/2} \delta \bar{k}_{nkpq}^{h_e} \tag{54}$$

$$\bar{\phi}_p^q = \frac{4\pi G}{g_R} \left[\frac{1}{2p+1} \bar{s}_p^q + \sum_{n=0}^{\infty} \sum_{k=-n}^{n} \frac{1}{2n+1} \bar{s}_n^k \delta \bar{k}_{nkpq}^{h_e} \right] . \tag{55}$$

$$\bar{s}_p^q = \frac{(2p+1) g_R}{4\pi G} \left[\bar{\phi}_p^q - \sum_{n=0}^{\infty} \sum_{k=-n}^{n} \bar{\phi}_n^k \delta \bar{k}_{nkpq}^{h_e} \right] . \tag{56}$$

$$s_n^k = \left[\frac{2(2n+1)}{1+\delta_k} \frac{(n-|k|)!}{(n+|k|)!} \right]^{1/2} \bar{s}_n^k \tag{57}$$

$$s = \sum_{n=0}^{\infty} \sum_{k=-n}^{n} \bar{s}_n^k \bar{\mathcal{H}}_n^k(\theta, \lambda) ; \tag{58}$$

$$\delta \bar{k}_{pq}^{h_e} \quad = \quad \left[(1+\epsilon)^{p+2} - 1 \right] \frac{2}{1+\delta_q} \frac{(p-|q|)!}{(p+|q|)!} \left[\frac{1+\delta_q}{2(2p+1)} \frac{(p+|q|)!}{(p-|q|)!} \right] \bar{\mathcal{H}}_p^q(\theta', \lambda')$$

$$= \frac{1}{2p+1}\left[(1+\epsilon)^{p+2}-1\right]\bar{\mathcal{H}}_p^q(\theta',\lambda') \tag{59}$$

$$\delta k_{mlpq}^{h_e} = \left[\frac{1+\delta_k}{2(2m+1)}\frac{(m+|l|)!}{(m-|l|)!}\right]^{1/2}\left[\frac{2(2p+1)}{1+\delta_k}\frac{(p-|q|)!}{(p+|q|)!}\right]^{1/2}\delta\bar{k}_{mlpq}^{h_e} \tag{60}$$

$$\begin{aligned}\delta\bar{k}_{pq}^{h_e} &= \sum_{m=0}^{\infty}\sum_{l=-m}^{m}\delta\bar{k}_{mlpq}^{h_e}\frac{2}{1+\delta_l}\frac{(m-|l|)!}{(m+|l|)!}\left[\frac{1+\delta_l}{2(2m+1)}\frac{(m+|l|)!}{(m-|l|)!}\right]\bar{\mathcal{H}}_m^l(\theta',\lambda')\\ &= \sum_{m=0}^{\infty}\sum_{l=-m}^{m}\frac{1}{2m+1}\delta\bar{k}_{mlpq}^{h_e}\bar{\mathcal{H}}_m^l(\theta',\lambda') \end{aligned}\tag{61}$$

$$\phi_0^{(e)} = Rg_R\sum_{n=0}^{\infty}\sum_{k=-n}^{n}\frac{4\pi G}{(2n+1)g_R}\bar{s}_n^k\left(\frac{R}{r}\right)^{n+1}\bar{\mathcal{H}}_n^k(\theta,\lambda) \tag{62}$$

$$\delta k_{nkpq}^{h_e} = \left[\frac{1+\delta_k}{2(2n+1)}\frac{(n+|k|)!}{(n-|k|)!}\right]^{1/2}\left[\frac{2(2p+1)}{1+\delta_k}\frac{(p-|q|)!}{(p+|q|)!}\right]^{1/2}\delta\bar{k}_{nkpq}^{h_e} \tag{63}$$

$$\delta\phi^{(e)} = Rg_R\sum_{n=0}^{\infty}\sum_{k=-n}^{n}\left[\frac{4\pi G}{(2n+1)g_R}\bar{s}_n^k\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\delta\bar{k}_{nkpq}^{h_e}\left(\frac{R}{r}\right)^{p+1}\bar{\mathcal{H}}_p^q(\theta,\lambda)\right], \tag{64}$$

$$\phi = Rg_R\sum_{p=0}^{\infty}\sum_{q=-p}^{p}\bar{\phi}_p^q\left(\frac{R}{r}\right)^{p+1}\bar{\mathcal{H}}_p^q(\theta,\lambda)\,. \tag{65}$$

$$\bar{\phi}_p^q = \frac{4\pi G}{g_R}\left[\frac{1}{2p+1}\bar{s}_p^q + \sum_{n=0}^{\infty}\sum_{k=-n}^{n}\frac{1}{2n+1}\bar{s}_n^k\delta\bar{k}_{nkpq}^{h_e}\right]. \tag{66}$$

$$\bar{s}_p^q = \frac{(2p+1)g_R}{4\pi G}\left[\bar{\phi}_p^q - \sum_{n=0}^{\infty}\sum_{k=-n}^{n}\bar{\phi}_n^k\delta\bar{k}_{nkpq}^{h_e}\right]. \tag{67}$$